

A Complete Lower-Sideband Boundary-Orbit Decomposition

Bin Wang

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Abstract

We complete the physical boundary-orbit extraction at the mandatory lower sideband $n_- = 2k - 2 = 2m$, where $m = k - 1$ is an orbit-period parameter on the same (σ, k) noise clock. Let

$$\Gamma_m = \{|f^j(p_{2m})| : 0 \leq j < 2m\}, \quad G_m = \frac{r_H^{-2m}}{1 + |M_m|}.$$

The folded orbit has exactly $2m$ distinct marked points and contributes the complete signed raw atom $-F_m^{\text{orb}}$, $F_m^{\text{orb}} = 2mG_m$. Its last critical point $\xi_m = h(p_{2m})$ lies in J^- or F according to $\epsilon_m = \mathbf{1}_{\{\xi_m \in J^-\}}$. The eventual cell counts are $(\epsilon_m, 0, 2m - \epsilon_m)$, and on a fixed physical phase

$$\frac{b - \xi_m}{\sqrt{\sigma}} \longrightarrow \frac{\sqrt{C_b}}{2u_c} \lambda^{1-\eta}.$$

The factor λ records that $m = k - 1$ while the noise clock remains k .

Removing the complete orbit gives the exact direct coefficient

$$p_{\sigma,k,2m} = \mathcal{T}_{k,m}^{\text{rest}} + \mathcal{P}_{\sigma,2m} - d_{\sigma,k,2m} - \mathcal{A}_{k,2m} - F_m^{\text{orb}}.$$

The point omitted by RH-339 is again super-target: $G_m/H_m = (C_M m)^{-1}(\beta R)^{2m}\{1+o(1)\} \rightarrow \infty$. The current period- $2k$ counterloop radial sideband has unknown sign but the sharp relative law

$$\frac{\mathcal{A}_{k,2m}}{F_m^{\text{orb}}} = \frac{C_M - 1}{m} + o(m^{-1}) \longrightarrow 0.$$

Thus the combined deterministic demand is eventually positive and asymptotic to the complete orbit atom, while neither its radial correction nor the orbit-free rest is proved target-negligible. We also obtain the shifted scalar interface $\mathcal{P}_{\sigma,2m}/F_m^{\text{orb}} \rightarrow C_* C_M \lambda^{\eta-1}$. No lower compensation, off-alias closure, determinant gluing, Gate progress, or Riemann-hypothesis conclusion follows.

1 One noise clock and the mandatory sideband

Work on the physical clock

$$k = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1), \quad \eta_\sigma = k - \frac{\log(1/\sigma)}{2 \log \lambda}, \quad R = \frac{7}{5}. \quad (1)$$

Set

$$m = k - 1, \quad n_- = 2m = 2k - 2, \quad H_m = mR^{-2m}. \quad (2)$$

The symbol m indexes a deterministic boundary orbit. It does not define a new noise sequence. Every one-alias cut $2k < h_\sigma \leq 4k$ contains n_- in its strict off-alias prefix [4].

Retain the same corrected folded noisy operator and the same frozen RH-334 cells

$$J^- = [0, 1] \cap [b - A\sqrt{\sigma}, b), \quad (3)$$

$$J^+ = [0, 1] \cap [b, b + A\sqrt{\sigma}], \quad (4)$$

$$F = [0, 1] \setminus (J^- \cup J^+). \quad (5)$$

For every order $n \geq 2$, the all-order physical identity is

$$q_{\sigma,k,n} = \mathcal{B}_{\sigma,k,n} + \mathcal{S}_{\sigma,k,n} + \mathcal{R}_{\sigma,k,n} + \mathcal{P}_{\sigma,n} - \mathcal{A}_{k,n}, \quad (6)$$

and the direct coefficient is

$$p_{\sigma,k,n} = q_{\sigma,k,n} - d_{\sigma,k,n}. \quad (7)$$

Here d is the separate noisy-head/counterloop defect. No full-trace/direct identification is assumed [6, 8].

At $n = 2m$, the current period- $2k$ counterloop contributes the exact radial sideband

$$\mathcal{A}_{k,2m} = 2(\beta^{2m} - \beta_k^{2m}), \quad \beta_k = \frac{|M_k|^{-1/(2k)}}{r_H}, \quad \beta = \frac{1}{r_H \sqrt{\lambda}}. \quad (8)$$

It is not an alias impulse of the period- $2k$ counterloop, and its sign is not fixed below [3].

2 Complete period- $2m$ orbit extraction

Let p_{2m} be the primitive boundary point and put

$$\Gamma_m = \{|f^j(p_{2m})| : 0 \leq j < 2m\}, \quad \xi_m = h(p_{2m}) < b. \quad (9)$$

RH-17 gives the ordered orbit and multiplier

$$M_m = (f^{2m})'(p_{2m}) = -C_M \lambda^m \{1 + o(1)\}, \quad C_M > 0, \quad (10)$$

and RH-339 proves that the other $2m - 1$ folded points lie in F eventually [1, 4].

Define

$$G_m = \frac{r_H^{-2m}}{1 + |M_m|}, \quad \epsilon_m = \mathbf{1}_{\{\xi_m \in J^-\}}. \quad (11)$$

Theorem 2.1 (Complete lower-sideband orbit atom). *For all sufficiently large k , Γ_m has exactly $2m$ distinct folded marked points with counts*

$$(\epsilon_m, 0, 2m - \epsilon_m) \quad (12)$$

in (J^-, J^+, F) . Define orbit-free raw rest slots by deleting Γ_m from each frozen cell. Then

$$\mathcal{B}_{\sigma,k,2m} = \mathcal{B}_{k,m}^{\text{rest}} - \epsilon_m G_m, \quad (13)$$

$$\mathcal{S}_{\sigma,k,2m} = \mathcal{S}_{k,m}^{\text{rest}}, \quad (14)$$

$$\mathcal{R}_{\sigma,k,2m} = \mathcal{R}_{k,m}^{\text{rest}} - (2m - \epsilon_m) G_m. \quad (15)$$

Thus, with

$$\mathcal{T}_{k,m}^{\text{rest}} = \mathcal{B}_{k,m}^{\text{rest}} + \mathcal{S}_{k,m}^{\text{rest}} + \mathcal{R}_{k,m}^{\text{rest}}, \quad F_m^{\text{orb}} = 2m G_m, \quad (16)$$

the exact aggregate raw identity is

$$\boxed{\mathcal{T}_{\sigma,2m} = \mathcal{T}_{k,m}^{\text{rest}} - F_m^{\text{orb}}.} \quad (17)$$

Proof. Primitivity and the RH-334 multiplier-preserving folding bijection give the cardinality and common weight. The retained $2m - 1$ points lie in F , while $\xi_m < b$ lies in J^- or F and never in J^+ . A finite set has zero multiplication operator on L^2 , so its localized noisy trace vanishes. Deleting its deterministic marked-point weights gives (13)–(15); summation proves (17). $\square$

The exact allocation is phase sensitive. Put

$$q_{b,m} = \frac{b - \xi_m}{\sqrt{\sigma}}. \quad (18)$$

The RH-327 inverse-branch expansion, applied to the period parameter m but the same σ , gives

$$q_{b,m} \longrightarrow \frac{\sqrt{C_b}}{2u_c} \lambda^{1-\eta} \quad (19)$$

along $\eta_\sigma \rightarrow \eta$. Indeed $C_b \lambda^{-2m}/\sigma \rightarrow C_b \lambda^{2-2\eta}$. Therefore $\epsilon_m = \mathbf{1}_{\{q_{b,m} \leq A\}}$ stabilizes away from equality of the limit with A ; at equality the source $o(1)$ term does not decide stabilization [5].

3 Exact typed lower coefficient

Theorem 3.1 (Complete lower-sideband coefficient). *At $n_- = 2m$,*

$$q_{\sigma,k,2m} = \mathcal{T}_{k,m}^{\text{rest}} + \mathcal{P}_{\sigma,2m} - \mathcal{A}_{k,2m} - F_m^{\text{orb}}, \quad (20)$$

$$p_{\sigma,k,2m} = \mathcal{T}_{k,m}^{\text{rest}} + \mathcal{P}_{\sigma,2m} - d_{\sigma,k,2m} - \mathcal{A}_{k,2m} - F_m^{\text{orb}}. \quad (21)$$

Consequently

$$p_{\sigma,k,2m} = o(H_m) \iff \mathcal{T}_{k,m}^{\text{rest}} + \mathcal{P}_{\sigma,2m} - d_{\sigma,k,2m} = \mathcal{A}_{k,2m} + F_m^{\text{orb}} + o(H_m). \quad (22)$$

The analogous full-trace statement omits $d_{\sigma,k,2m}$.

Proof. Substitute (17) into (6), then use (7). Rearrangement proves the equivalence. $\square$

Every direct prefix containing $2m$ has the nonnegative summand

$$\frac{|p_{\sigma,k,2m}| R^{2m}}{2m} = \frac{|p_{\sigma,k,2m}|}{2H_m}. \quad (23)$$

Thus direct prefix closure implies (22). Likewise RH-339's off-alias full-trace prefix forces the q version. Neither converse controls the remaining orders.

If $D_m^{\text{orb}} = (2m - 1)G_m$ is the RH-339 partial atom and $C_k^- = q_{\sigma,k,2m} + D_m^{\text{orb}}$, then exactly

$$C_k^- = \mathcal{T}_{k,m}^{\text{rest}} + \mathcal{P}_{\sigma,2m} - \mathcal{A}_{k,2m} - G_m. \quad (24)$$

This physically expands the combined complement used in the earlier necessary compensation law.

4 Missing point and radial sideband scales

Theorem 4.1 (Complete atom and radial relative law). *As $m = k - 1 \rightarrow \infty$,*

$$F_m^{\text{orb}} = \frac{2m}{C_M} \beta^{2m} \{1 + o(1)\}, \quad (25)$$

$$\frac{F_m^{\text{orb}}}{H_m} = \frac{2}{C_M} (\beta R)^{2m} \{1 + o(1)\} \longrightarrow \infty. \quad (26)$$

Moreover

$$\frac{F_m^{\text{orb}}}{D_m^{\text{orb}}} = \frac{2m}{2m - 1}, \quad F_m^{\text{orb}} - D_m^{\text{orb}} = G_m, \quad (27)$$

and

$$\frac{G_m}{H_m} = \frac{1}{C_M m} (\beta R)^{2m} \{1 + o(1)\} \longrightarrow \infty. \quad (28)$$

The radial counterloop term satisfies the sharp relative law

$$\boxed{\frac{\mathcal{A}_{k,2m}}{F_m^{\text{orb}}} = \frac{C_M - 1}{m} + o(m^{-1}) \longrightarrow 0.} \quad (29)$$

Consequently

$$S_m^- := F_m^{\text{orb}} + \mathcal{A}_{k,2m} > 0 \quad \text{eventually,} \quad \frac{S_m^-}{F_m^{\text{orb}}} \longrightarrow 1. \quad (30)$$

Proof. The multiplier law (10) gives $G_m = C_M^{-1} \beta^{2m} \{1 + o(1)\}$, proving (25)–(28); RH-336 supplies the exact $\beta R > 1$ certificate [7].

For the current period- $2k$ counterloop,

$$\beta_k = \beta \exp \left[-\frac{\log C_M}{2k} + o(k^{-1}) \right]. \quad (31)$$

Since $m/k \rightarrow 1$,

$$\frac{\beta_k^{2m}}{\beta^{2m}} = \exp \left\{ -\frac{m}{k} \log C_M + o(1) \right\} \longrightarrow C_M^{-1}.$$

Insert this into (8):

$$\mathcal{A}_{k,2m} = 2\beta^{2m} \{1 - C_M^{-1} + o(1)\}.$$

Division by (25) proves (29). Equation (30) follows without any sign assumption on the radial term. $\square$

The radial sideband is small only relative to the full orbit atom. Because $F_m^{\text{orb}}/H_m \rightarrow \infty$, (29) does not prove $\mathcal{A}_{k,2m} = o(H_m)$. Its sign and target scale remain source-unresolved. The missing point G_m , by contrast, is definitely super-target.

5 Shifted lower parity interface

The actual square-root parity law gives, since $m\sqrt{\sigma} \rightarrow 0$,

$$\mathcal{P}_{\sigma,2m} = 2mC_*\sqrt{\sigma} r_H^{-2m} \{1 + o(1)\}. \quad (32)$$

Corollary 5.1 (Lower-sideband phase ratio). *Along $\eta_\sigma \rightarrow \eta$,*

$$\frac{\mathcal{P}_{\sigma,2m}}{F_m^{\text{orb}}} \longrightarrow C_* C_M \lambda^{\eta-1}, \quad \frac{\mathcal{P}_{\sigma,2m}}{S_m^-} \longrightarrow C_* C_M \lambda^{\eta-1}. \quad (33)$$

Proof. Divide (32) by (25). The remaining clock factor is $\sqrt{\sigma} \lambda^m = \lambda^{\eta_\sigma - k + m} = \lambda^{\eta_\sigma - 1}$. The second limit uses (30). $\square$

Thus the unique symbolic scalar balance for RH-347 is

$$\eta_- = 1 - \frac{\log(C_* C_M)}{\log \lambda}. \quad (34)$$

This is an interface, not a compensation result: the signed orbit-free remainder in (21) is still unestimated.

6 Executable audit and claim boundary

The artifact reconstructs the lower boundary orbit at $k = 3, 5, 9, 17, 33$, checks the complete counts and closure, evaluates the missing point, radial sideband, and shifted parity ratios, and verifies an independent rational typed ledger. Finite decimals, including the observed radial sign, are reproduction checks only and are not interval certificates or asymptotic evidence.

RH-346 proves the complete physical lower orbit subledger, exact coefficient decomposition, super-target missing point, sharp radial/full relative law, and shifted parity interface. It proves neither lower compensation nor lower noncompensation. One sideband does not close the punctured off-alias aggregate, and no head transport or direct prefix theorem follows.

RH-347 may audit the scalar mechanism only with the exact combined demand S_m^- ; RH-345 is not copied with a changed label [2]. RH-288 remains inactive. Gates A–E remain false/open. This paper constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt prime-power trace, proves no completed-zeta divisor equality, and does not prove the Riemann Hypothesis.

References

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